IMPERIAL RESEARCH ARM · KAGGLE QUALIFIER

Name That LLM

Nabil Shah · MSc Advanced Computing @ Imperial College London · October 2026

THE TASK

Which of 18 language models wrote this passage?

18
models, six labs

2,699
training passages

4,000
test passages, unseen prompts

BA
balanced accuracy

Hardest: glm-5.3 vs glm-5.3-flash, gemini-3.7 vs 3.8-flash.

LITERATURE REVIEW

Research before code.

22
public notebooks read

18
discussion threads

15
research notes, each ending in a decision

~100
sources cited

Data beats size

A bigger pool: 0.569 → 0.611.

The spine is public

0.620 blend, weights published.

No public OOF

So validate on my own folds.

STARTING POINT

Validation first, then build on published work.

01

Freeze the folds

Grouped by prompt, like the hidden test.

02

Plain baseline

n-gram SVC, host data only: **0.348**

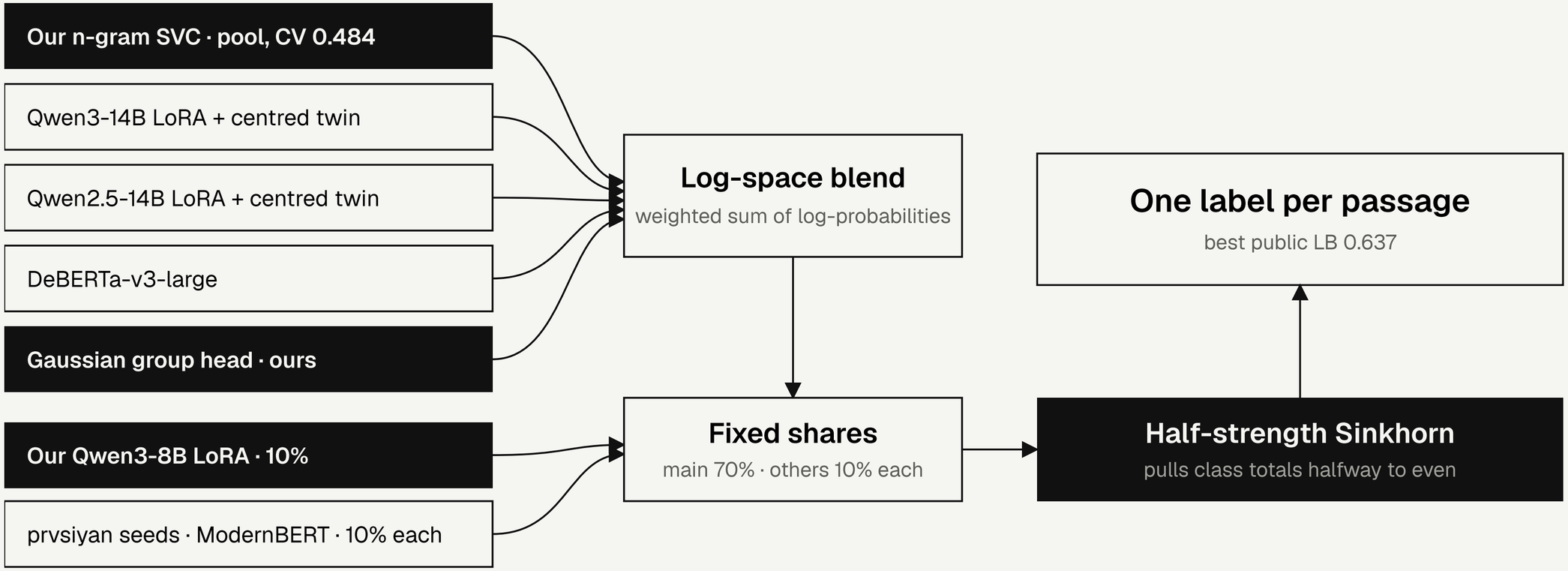
03

Read, don't fork

Our own loader for the best public weights (0.620).

APPROACH

Many views, one blend, then balancing.



Filled = ours. Outlined = published, through our own loader.

HYPOTHESES

What would separate the models?

Style lives in small habits	n-gram SVC: 0.348
Data beats model size	+ 84k public passages: **0.484**
The prompt hides the author	Gaussian group head: **+0.058**
The test mix is uneven	Half-strength balancing: **+0.003**

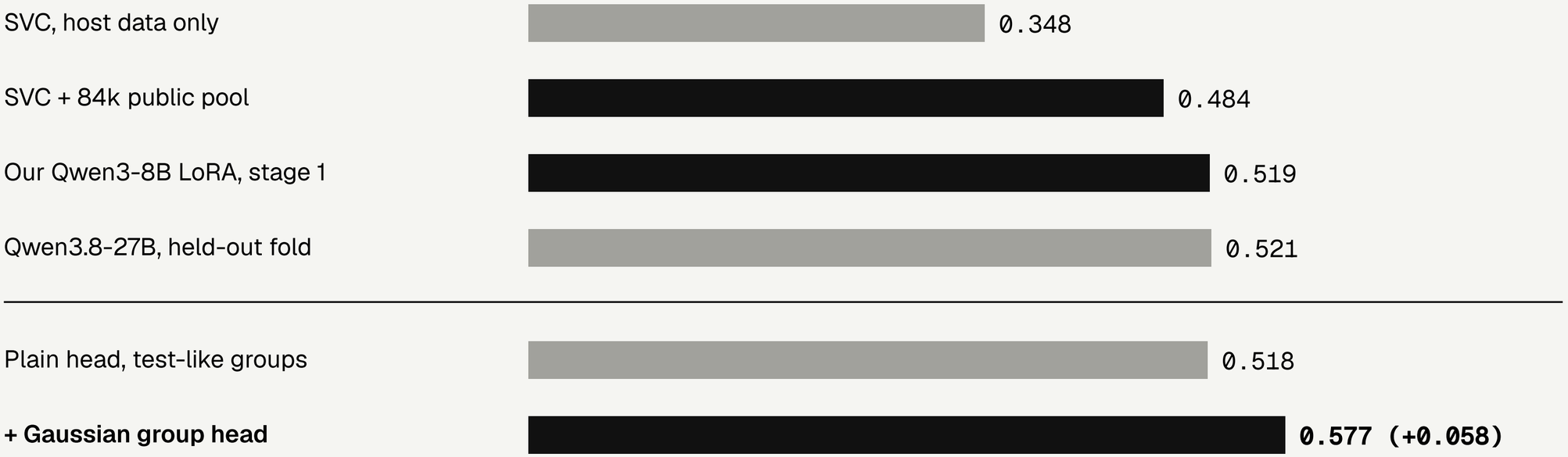
ABLATION · PUBLIC LEADERBOARD

One change per probe. Final picks: 9 and 7.

PROBE	CHANGE	PUBLIC LB	Δ
1	Our SVC + published spine	0.626	—
2	Half-strength balancing	0.629	**+0.003**
3	+ two public Qwen3-8B seeds	0.631	+0.002
4	+ in-style Qwen3-8B	0.627	−0.004
5	+ UltraBERT, ATLAS	0.631	±0
6	Peer correction	0.622	−0.009
7	+ our own Qwen3-8B LoRA	0.631	±0
8	+ Gaussian head (weight 0.4)	0.631	±0
9	Gaussian head at 1.0 + ModernBERT	**0.637**	**+0.006**
10	+ Gaussian heads on both 14B LoRAs	0.630	−0.007

ABLATION · HONEST CROSS-VALIDATION

Honest scores, on unseen folds.



Within noise: centred SVC, pair specialists, bigger SVC pool.

WHAT DIDN'T WORK

Peer correction: 0.631 → 0.622

The idea

Remove a topic's shared lean. Drop class quotas.

Why we tried it

+0.010 for its author, on a weaker blend.

Why it failed

No quotas: a strong blend drifts to its favourites.

WHAT THE LEADERBOARD TOLD US

Four ties at 0.631, then 0.637.

Small gains round away

So we steered by cross-validation.

Honest features matter

The head helped at full weight on our own features, and hurt on the 14B's.

0.637 best public score · from 0.620

HOW I USED AI

A repo built for agents from the first hour.

01

Structure

AGENTS.md and CLAUDE.md, with the rules as invariants. Phases and tickets.

02

Method

Research note, then a recommendation. One log row per change.

03

Reproducible

Frozen folds, tests first, seeds fixed, every model run on Kaggle.

04

Parallel

Six side sessions, each in its own branch, while I made the calls.

My rule: one change per probe, kept only if CV or a research note backs it. Every session in ai-sessions/

WHAT I'D TRY NEXT

More data, honest numbers everywhere.

01

Bigger pool

Retrain on 122k open passages.

02

Five folds each

Fit the blend on honest scores.

03

Joint decoding

One Gaussian model for all members.

04

Hard pairs

Targeted data for the siblings.

github.com/Nabzx/name-that-llm · Thanks for watching.